

Lecture 5

Topics: Combinational Logic Applications



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What is a Combinational Circuit?

- A **combinational circuit** is a type of digital circuit where the output depends solely on the current inputs, not on past inputs or states. These circuits are built using logic gates (AND, OR, NOT, etc.) and perform specific logical operations, like addition, subtraction, and data routing.
- **Key Characteristics:**
 - Outputs are a direct function of inputs.
 - No memory or feedback loops are involved.

Combinational Logic Applications

- Full adder
 - Seven-Segment Displays
 - Decoders
 - Multiplexers
 - Demultiplexers
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- Reading: chapter 8
 - https://faculty.etsu.edu/tarnoff/ntes2150/Ch8_v02.pdf

Seven-Segment Displays

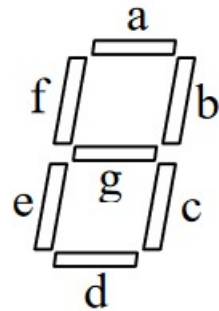


Figure 8-9 7-Segment Display

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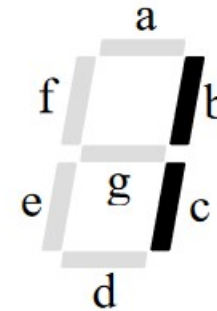


Figure 8-10 Displaying a '1'

Figure 8-10 Displaying a '1'

It is the most common way to display time on a clock radio, and it is one of the easiest ways to implement a numeric output for a digital circuit.

Seven-Segment Displays (Cont.)

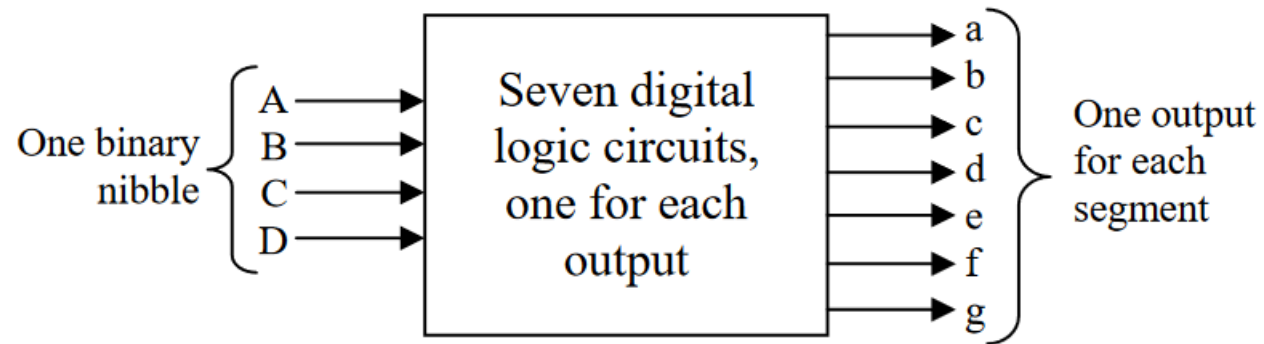


Figure 8-12 Block Diagram of a Seven-Segment Display Driver

Seven-Segment Displays (Cont.)

	Digit	Segments		Digit	Segments
	0	a, b, c, d, e, f		1	b, c
	2	a, b, d, e, g		3	a, b, c, d, g
	4	b, c, f, g		5	a, c, d, f, g
	6	a, c, d, e, f, g		7	a, b, c
	8	a, b, c, d, e, f, g		9	a, b, c, d, f, g
	A	a, b, c, e, f, g		B	c, d, e, f, g
	C	a, d, e, f		D	b, c, d, e, g
	E	a, d, e, f, g		F	a, e, f, g

Figure 8-13 Segment Patterns for all Hexadecimal Digits

Figure 8-13 Segment Patterns for all Hexadecimal Digits

Seven-Segment Displays (Cont.)

Inputs				Hex Value	Segments						
A	B	C	D		a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	1	0	1	1	0	0	0	0
0	0	1	0	2	1	1	0	1	1	0	1
0	0	1	1	3	1	1	1	1	0	0	1
0	1	0	0	4	0	1	1	0	0	1	1
0	1	0	1	5	1	0	1	1	0	1	1
0	1	1	0	6	1	0	1	1	1	1	1
0	1	1	1	7	1	1	1	0	0	0	0
1	0	0	0	8	1	1	1	1	1	1	1
1	0	0	1	9	1	1	1	1	0	1	1
1	0	1	0	A	1	1	1	0	1	1	1
1	0	1	1	B	0	0	1	1	1	1	1
1	1	0	0	C	1	0	0	1	1	1	0
1	1	0	1	D	0	1	1	1	1	0	1
1	1	1	0	E	1	0	0	1	1	1	1
1	1	1	1	F	1	0	0	0	1	1	1

Figure 8-14 Seven Segment Display Truth Table

Figure 8-14 Seven Segment Display Truth Table

Seven-Segment Displays (Cont.)

AB \ CD				
	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	1	1	1	1
10	1	0	1	1

Figure 8-15 Karnaugh Map for Segment 'e'

Figure 8-15 Karnaugh Map for Segment 'e'

$$\text{Segment } e = (\bar{B} \cdot \bar{D}) + (A \cdot B) + (A \cdot C) + (C \cdot \bar{D})$$

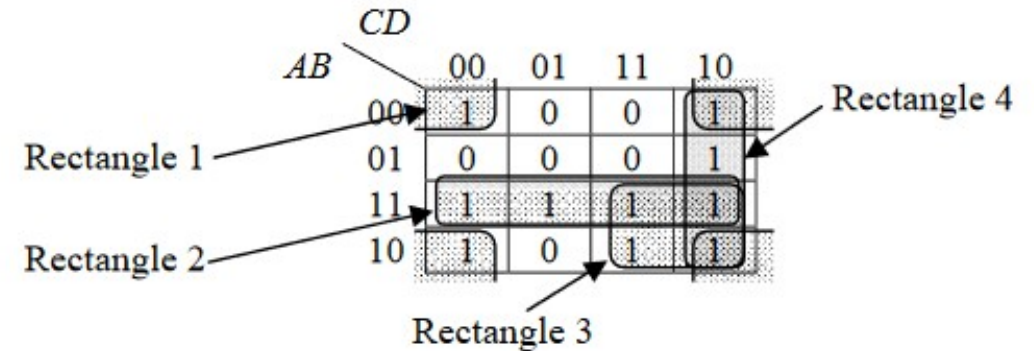


Figure 8-16 Karnaugh Map for Segment 'e' with Rectangles

Figure 8-16 Karnaugh Map for Segment 'e' with Rectangles

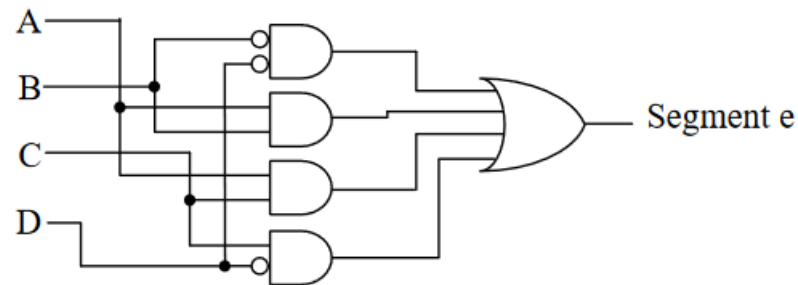
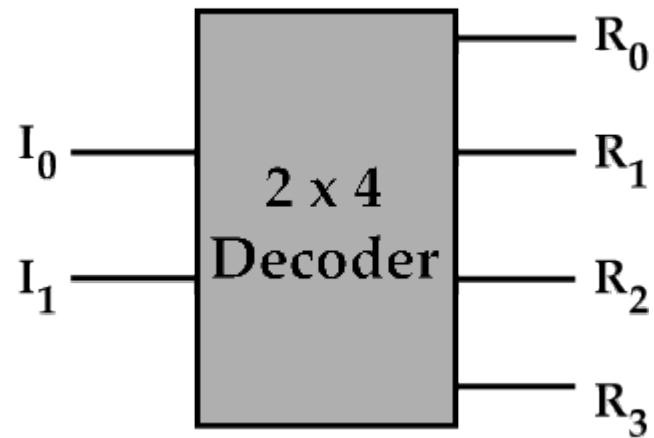


Figure 8-17 Logic Circuit for Segment e of 7-Segment Display

Figure 8-17 Logic Circuit for Segment e of 7-Segment Display

Decoders

- A decoder in digital electronics is a combinational circuit that converts a binary number into a specific output pattern. Essentially, **a decoder takes a binary input value and activates one particular output, turning it "on,"** while keeping other outputs "off." Decoders are widely used in various applications, especially in memory systems and for the selection of particular devices in a system.



Decoder – an analogy

- Think of a decoder as a floor selector in a hotel elevator. Each button in the elevator corresponds to a floor. If you press the 3rd-floor button (binary input), the elevator stops only on the 3rd floor (one output is activated).

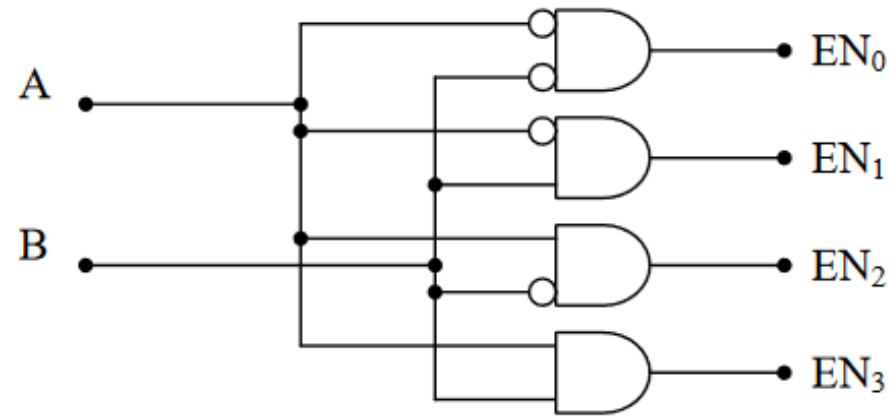
Decoder

- For an n -bit input, there are 2^n possible combinations. An n -to- 2^n decoder will have n inputs and 2^n outputs. Based on the input, only one of the 2^n outputs will be active.

Truth Table for an 2-to-4 Decoder

Inputs		Outputs				X
A	B	d ₀	d ₁	d ₂	d ₃	
0	0	1	0	0	0	0
0	1	0	1	0	0	1
1	0	0	0	1	0	2
1	1	0	0	0	1	3

Digital Circuit for a 2-to-4 Decoder



Decoder - Applications

- **Memory Addressing:** Decoders are often used in memory systems to decode the address lines to select a specific memory location.
- **Device Selection:** In a system with multiple devices, a decoder can determine which device to activate based on the given binary input.
- **Display Systems:** 7-segment display decoders are used to convert a binary number to the appropriate pattern for displaying numbers on a 7-segment display.

Multiplexers

- A multiplexer (often abbreviated as MUX) is a combinational digital circuit designed to **select one of many inputs and forward the selected input to a single output line**. The selection is determined by a set of selection lines. Think of a multiplexer as a multi-input, single-output switch¹

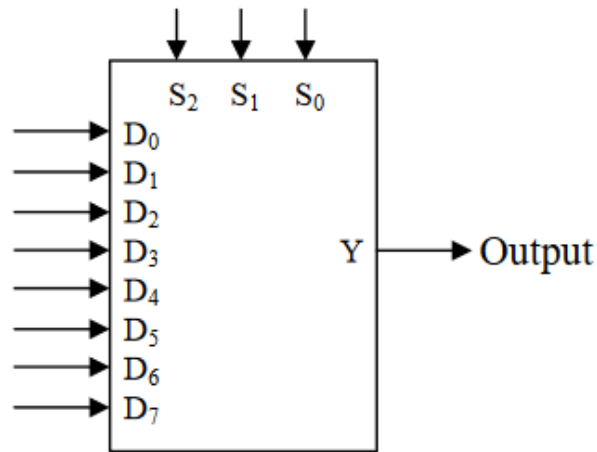


Figure 8-25 Block Diagram of an Eight Channel Multiplexer

Multiplexers – an analogy

- Picture a TV remote control. The remote (selection lines) lets you choose between different channels (inputs), but only one channel is displayed on the screen (output) at a time.

Truth Table for an Eight Channel Multiplexer

S ₂	S ₁	S ₀	Y
0	0	0	D ₀
0	0	1	D ₁
0	1	0	D ₂
0	1	1	D ₃
1	0	0	D ₄
1	0	1	D ₅
1	1	0	D ₆
1	1	1	D ₇

Figure 8-26 Truth Table for an Eight Channel Multiplexer

if the selector inputs are set to $S_2 = 0$, $S_1 = 1$, and $S_0 = 1$, then the data present at D3 will be output to Y. If D3 = 0, then Y will output a 0.

Multiplexer : Applications

- **Data Routing:** MUXes can be used to route data from multiple sources to a single destination.
- **Function Implementation:** With the appropriate data inputs, a multiplexer can be used to implement any Boolean function.
- **Resource Sharing:** Allow multiple devices to share a single resource, like a data bus.

Demultiplexers

- Demultiplexers take a single input and select one channel out of a group of output channels to which it will route the input

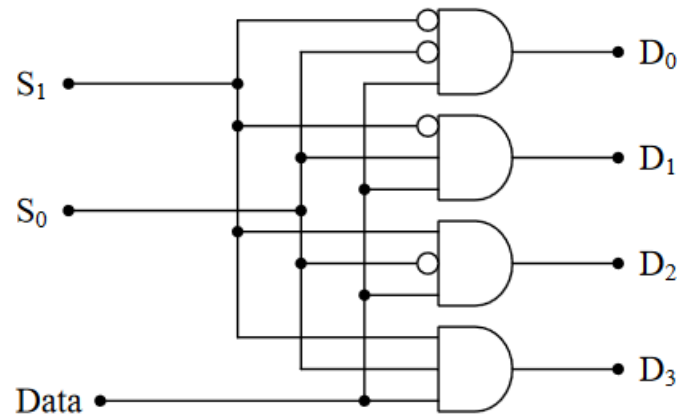


Figure 8-27 Logic Circuit for a 1-Line-to-4-Line Demultiplexer

if $S_1=0$ and $S_0=1$, then the D_1 output will match the input from the Data line and outputs D_0 , D_2 , and D_3 will be forced to have an output of zero. If $S_1=0$, $S_0=1$, and Data=0, then $D_1=0$. If $S_1=0$, $S_0=1$, and Data=1, then $D_1=1$.

Demultiplexers – an analogy

- A demultiplexer is like a post office sorting machine. It receives one mail (input) and, based on the address (control signal), sends it to the right street or home (output).

Truth Table for a 1-Line-to-4-Line Demultiplexer



The diagram shows a 1-Line-to-4-Line Demultiplexer. It has two select inputs, S₁ and S₀, and one data input, Data. There are four data outputs, D₀, D₁, D₂, and D₃. The truth table below defines the output values for each combination of inputs.

S ₁	S ₀	Data	D ₀	D ₁	D ₂	D ₃
0	0	0	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	1	0	0
1	0	0	0	0	0	0
1	0	1	0	0	1	0
1	1	0	0	0	0	0
1	1	1	0	0	0	1

Figure 8-28 Truth Table for a 1-Line-to-4-Line Demultiplexer

Demultiplexer: Applications

- 1.Data Routing:** A primary application of demultiplexers is to route data from a single source to multiple destinations, depending on the control signal. For instance, in a communication system, a demux can direct incoming data to one of several possible devices or channels.
- 2.Display Driving:** Demuxes are often used in display systems where, based on the control signal, a specific segment or part of the display is activated. For example, driving a 7-segment display might use demultiplexers to select which segment to activate.
- 3.Memory Addressing:** In memory systems, a demultiplexer can be used to select a specific memory location (or set of locations) among many. It can decode a binary address to a specific memory cell selection.

Next lecture

- Memory cells or Latches
- The SR and D Latch
- Sequential circuit
 - Divide-By-Two Circuit
 - Counter